Cocoa Booms and Human Capital in Ghana: Evidence from Census Microdata

APEP Autonomous Research*

February 24, 2026

Abstract

How do commodity booms affect human capital in developing countries? We study the 2000s cocoa price boom using three rounds of Ghana's population census (1984, 2000, 2010), comprising 5.7 million observations. Employing a Bartik-style specification and a doubly-robust difference-in-differences estimator, we compare outcomes in high- versus low-cocoa-intensity regions within Ghana's forest belt. Adult employment fell by 6.8 percentage points per unit of cocoa share—robust to exact randomization inference across all 720 permutations of regional treatment assignments (RI $p = 0.014$). Literacy gains of 5.5 percentage points appear under conventional clustered inference but are less certain under exact permutation tests, underscoring inference challenges with few clusters. Pre-trend tests using the 1984–2000 period show zero differential trends. Our findings suggest that commodity booms can reshape labor markets and improve human capital in developing countries.

JEL Codes: I25, O12, O13, Q17

Keywords: cocoa, commodity boom, human capital, education, child labor, Ghana, doubly-robust

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1. Introduction

In October 2000, a metric ton of cocoa sold for $906 on world markets. By 2010, that same ton fetched $2,800—a 209% increase in nominal terms. For Ghana, the world's second-largest cocoa producer and a country where roughly 800,000 farming households depend on the crop, this price shock represented a first-order change in economic conditions (**?**). Whether such windfalls build or erode human capital is among the most consequential questions in development economics, and the answer is far from settled.

The theoretical ambiguity is well known. Higher commodity prices raise household income, relaxing credit constraints that keep children out of school and enabling investments in educational quality (**?**). But they simultaneously raise the opportunity cost of schooling by increasing the returns to agricultural labor, potentially pulling children into the fields at the expense of the classroom (**?**). Which force dominates depends on the structure of production, the age and gender of affected children, the degree of credit constraints, and the time horizon of the boom. In Ghana's cocoa sector—where the International Labour Organization estimates that over 700,000 children were engaged in hazardous work as recently as 2020—these are not merely academic distinctions (**?**).

We study the effect of the 2000s cocoa boom on human capital accumulation in Ghana using individual-level census microdata from IPUMS International. Our dataset comprises three rounds of the Ghana Population and Housing Census—1984, 2000, and 2010—totaling 5.7 million person records with regional geographic identifiers. This represents, to our knowledge, the first use of census microdata to examine the human capital consequences of cocoa price shocks in a major producing country.

Our identification strategy exploits the sharp geographic concentration of cocoa production within Ghana's forest belt. Four regions—Western, Ashanti, Brong Ahafo, and Eastern—account for over 95% of national cocoa production (**?**). We compare changes in education and employment outcomes between high- and low-cocoa-intensity regions, restricting our main specification to the six ecologically similar forest-belt regions to ensure that treated and control areas share comparable climate, agricultural structure, and pre-existing development trajectories. The key exogenous variation comes from world cocoa prices, which are determined on global commodity markets and are plausibly exogenous to conditions in any single Ghanaian region.

We implement a doubly-robust difference-in-differences (DR DiD) estimator following **?**. This approach combines an outcome regression model with an inverse probability weighting model, yielding estimates that are consistent if either model is correctly specified—a valuable property given that cocoa-producing regions differ from non-producing regions on multiple

observable dimensions. We complement the DR estimates with standard two-way fixed effects regressions using continuous treatment intensity (the Bartik interaction of regional cocoa shares with world prices).

Three sets of results emerge. First, the boom *reduced* adult employment in cocoa regions—a finding that is robust across all inference approaches. Employment rates fell by 6.8 percentage points per unit of cocoa share, confirmed by exact randomization inference (RI $p = 0.014$) and stable across leave-one-region-out specifications. The DR DiD confirms a 3.1 percentage point decline in high-cocoa regions ($p < 0.001$). Agricultural employment declined by similar magnitudes. This pattern is consistent with a structural transformation channel: higher cocoa incomes enabled households to reduce labor supply, with workers exiting agriculture for non-farm activities or reducing participation altogether.

Second, the cocoa boom *improved* educational outcomes for school-age children (ages 6–17) in high-cocoa regions, though the precision of this finding depends on the inference approach. The continuous-treatment specification shows literacy rose by 5.5 percentage points per unit of cocoa share (conventional clustered $p < 0.05$), and the DR DiD estimates a 3.1 percentage point gain ($p < 0.001$). However, exact randomization inference across all 720 permutations yields an RI p-value of 0.34 for literacy, illustrating the challenge of conducting inference with only 6 clusters (**?**). Leave-one-region-out analysis shows the literacy coefficient is positive in all specifications (range: 0.041–0.103), suggesting the sign is reliable even if the conventional significance level is overstated.

Third, the pre-trend tests strongly support our identification strategy. Using the 1984–2000 period—when world cocoa prices were declining—we estimate the same specification and find treatment coefficients that are economically tiny and statistically indistinguishable from zero for enrollment ($\beta = 0.0005$, p = 0.98), primary completion (β = -0.0001, p = 0.997), employment ($\beta = 0.009$, p = 0.60), and agricultural employment ($\beta = 0.018$, p = 0.86). Literacy cannot be tested in the 1984–2000 pre-trend because the LIT variable is unavailable in the 1984 census, though the closely related outcomes of enrollment and primary completion show zero pre-trends. The three-period event study shows a clear break: null effects in the pre-boom period, followed by significant effects that emerge only during the boom.

Our paper contributes to several literatures, which we discuss in detail in the next section. Most directly, we add to the growing body of work on commodity price shocks and human capital in developing countries (**?????**). Our contribution is to provide the first individual-level census evidence for Ghana—the world's second-largest cocoa producer—using a rigorous DR DiD framework with three census periods that enable pre-trend testing. We also speak to the broader resource economics literature on whether commodity booms are "curses" or "blessings" for development (**??**), and we contribute methodologically by demonstrating

the value of doubly-robust estimation (**?**) in a development setting with few clusters and substantial compositional differences between treated and control areas.

The remainder of the paper proceeds as follows. Section **??** reviews the related literature. Section **??** describes the institutional background of cocoa production in Ghana and the 2000s price boom. Section **??** presents a conceptual framework. Section **??** describes our data. Section **??** details the empirical strategy. Section **??** presents results. Section **??** discusses robustness and threats to validity. Section **??** concludes.

2. Related Literature

Our paper sits at the intersection of four literatures: commodity price shocks and human capital, child labor and schooling in developing countries, the resource curse and structural transformation, and methodological advances in difference-in-differences estimation.

2.1 Commodity Prices and Human Capital

Commodity price swings force a choice between the classroom and the field, and the evidence on which wins is strikingly mixed. The key theoretical tension is between income effects—which relax credit constraints on schooling—and substitution effects—which raise the opportunity cost of time in school.

? provides the canonical finding that substitution effects dominate: coffee booms in Brazil reduced school enrollment among children aged 14–17, who were most productive in coffee harvesting. **?** study the opposite shock—the 1990 cocoa price collapse in Côte d'Ivoire—and find that the crisis reduced schooling, particularly for children of cocoa farmers, suggesting income effects were important in that context. The asymmetry between these findings hints at heterogeneity across crops, production structures, and initial conditions.

In Vietnam, **?** documents that rice price liberalization raised school enrollment for children under 15, consistent with income effects dominating in a setting where rice is produced by smallholder households. **?** provides complementary evidence that child labor declines as economic status improves, supporting the view that poverty drives child labor rather than labor demand alone. **?** exploits agricultural shocks in Tanzania to show that negative income shocks increase child labor and reduce schooling, providing a symmetric piece of evidence: if bad shocks hurt schooling, good shocks should help.

Beyond agriculture, **?** study the Appalachian coal boom and find that tight labor markets reduced educational attainment by raising the opportunity cost of schooling—a substitution effect operating through adult labor demand rather than child labor. **?** find a similar pattern for U.S. housing booms, and **?** documents that the expansion of export manufacturing in

Mexico reduced schooling for those with newly available factory jobs. These studies from middle- and high-income settings consistently find substitution effects dominating, suggesting that income effects may be more important in low-income settings where credit constraints on schooling are more binding.

Our setting—smallholder cocoa farming in Ghana with high baseline poverty and significant school fees—is precisely where theory predicts income effects should dominate. The tree-crop production structure of cocoa, which limits short-run labor intensification, further weakens the substitution channel. We find that income effects do dominate, consistent with the broader pattern identified by **?** in his survey: child labor is primarily a poverty phenomenon.

2.2 Income Shocks, Schooling, and Child Labor

A related strand of the literature examines how household income shocks—not necessarily through commodity prices—affect children's schooling. **?** show that early-life rainfall shocks in Indonesia have persistent effects on women's educational attainment, health, and socioeconomic status. **?** provides causal evidence that droughts in India reduce human capital accumulation both contemporaneously (through reduced schooling) and in the long run (through lower adult earnings), with the effects concentrated among the poorest households.

? study conditional cash transfers in Ecuador and find that transfers improve child development, particularly for the poorest families. **?** document similar effects in Cambodia. **?** demonstrates that school construction in Indonesia—which reduced the cost of schooling rather than raising income—had large effects on educational attainment and earnings, establishing the sensitivity of schooling decisions to economic conditions in developing countries.

The theoretical foundation for understanding these income-schooling links comes from **?**, who model child labor as a consequence of household poverty: families send children to work only when adult wages fall below a subsistence threshold. Their "luxury axiom"—that child labor is a bad that households consume less of as they become richer—predicts that positive income shocks should reduce child labor and increase schooling. Our results are consistent with this prediction: the cocoa boom raised household incomes above the threshold where child labor was necessary, enabling greater investment in education.

2.3 Resource Booms and Development

Whether natural resource windfalls promote or hinder development is one of the oldest questions in development economics. **?** document a negative cross-country correlation between resource abundance and growth—the "resource curse"—which **?** attributes to Dutch disease, rent-seeking, and institutional deterioration. The theoretical mechanism, formalized

by **?**, operates through factor reallocation: resource booms draw labor and capital into the booming sector and away from manufacturing, potentially reducing long-run growth.

Recent microeconomic evidence paints a more nuanced picture. **?** study U.S. oil and gas booms and find positive effects on local wages and employment, with no evidence of Dutch disease in manufacturing. **?** find that Brazilian oil windfalls improve local living standards, though the gains are concentrated among public sector workers. **?** document positive spillovers from a Peruvian gold mine to surrounding communities. **?** find heterogeneous effects of commodity prices on conflict in Colombia—oil price increases reduce violence, while agricultural price increases intensify it—suggesting the institutional and economic structure of the commodity sector matters critically.

Our paper speaks to this literature by examining human capital—a key channel through which resource booms could affect long-run development. If commodity windfalls increase schooling, they may lay the foundation for sustained growth even after prices decline. The structural transformation we document—workers exiting agriculture for other sectors—is consistent with the classic models of **?** and **?**, with **?**'s evidence that structural change has been growth-enhancing in Africa since 2000, and with **?**'s analysis of commodity-driven urbanization in Africa.

2.4 Econometric Methods

We employ the doubly-robust difference-in-differences estimator of **?**, which nests the standard DiD estimator as a special case but is robust to misspecification of either the outcome model or the propensity score model. This estimator belongs to the growing family of methods that address limitations of conventional two-way fixed effects (**??**). While our setting has only two periods (2000–2010) for the main treatment comparison, making the concerns about staggered treatment timing in **?** less relevant, the DR approach is valuable because cocoa-producing regions differ substantially from non-producing regions on observable characteristics.

Our Bartik-style continuous-treatment specification follows **?**, who provide conditions under which shift-share instruments yield consistent estimates. The theoretical foundations for inference in shift-share designs are developed by **?** and **?**. In our case, the "shift" (world cocoa price change) is plausibly exogenous to conditions in any single Ghanaian region, while the "shares" (regional cocoa production shares) reflect pre-treatment geographic specialization. The key identifying assumption is that cocoa-intensive regions would have experienced parallel trends in the absence of the price boom—an assumption we support with the 1984 census pre-trend test. We follow **?**'s caution that pre-tests have limited power and complement them with the formal sensitivity analysis of **?**.

Given the small number of clusters (6 in the forest-belt specification), conventional

cluster-robust standard errors may overreject (**??**). We address this in three ways. First, we complement the Bartik regressions with the DR DiD estimator, whose analytical standard errors based on the influence function are less sensitive to the number of clusters. Second, we conduct *exact randomization inference* (RI): with 6 forest-belt regions, we enumerate all $6! = 720$ permutations of cocoa share assignments and compute the fraction of permuted coefficients exceeding the observed value in absolute terms. This yields finite-sample-exact p-values that are valid regardless of cluster count. Third, we report leave-one-region-out estimates to verify that no single region drives our results. The TWFE decomposition of **?** is not directly relevant to our setting (two periods, no staggering), but we note that the DR DiD estimator provides robustness to treatment effect heterogeneity. The semiparametric DiD estimator of **?** provides an additional methodological foundation for our approach to covariate adjustment.

3. Institutional Background

3.1 Cocoa in Ghana's Economy

Ghana has been a major cocoa producer since the late nineteenth century, and the crop remains central to the national economy (**?**). In the early 2000s, cocoa accounted for approximately 30% of export earnings, 15% of government revenue, and 7–8% of GDP (**?**). An estimated 800,000 farming households—roughly six million people—depended directly on cocoa cultivation, with millions more engaged in downstream processing, transport, and trade.

Cocoa production in Ghana is geographically concentrated in the country's southern forest belt, reflecting the crop's ecological requirements of high rainfall (1,200–3,000 mm annually), tropical temperatures, and humid conditions (**?**). Four regions dominate: Western Region produces roughly 55% of national output, Ashanti 20%, Brong Ahafo 12%, and Eastern 8%. Together, these four regions account for over 95% of Ghana's cocoa. Central and Volta Regions produce small quantities, while Greater Accra (urbanized) and the three northern savanna regions (Northern, Upper East, Upper West) produce essentially none.

3.2 The COCOBOD Price-Setting Mechanism

A distinctive feature of Ghana's cocoa sector is the government's active role in price setting through the Ghana Cocoa Board (COCOBOD). Established in 1947, COCOBOD purchases all cocoa from farmers at a government-determined farmgate price, which is announced before each main crop season. COCOBOD then exports the cocoa at world market prices, using the

margin between the farmgate and export prices to fund road maintenance, disease control, input subsidies, and agricultural extension services.

The farmgate price partially insulates farmers from world price volatility, but transmission is far from zero. The government adjusts the farmgate price in response to world price movements, typically passing through 50–70% of world price changes with a lag of one to two seasons (**?**). During the 2000s boom, the farmgate price rose from approximately 250 cedis per 64-kg bag in 2000 to 3,200 cedis by 2010—reflecting both world price increases and domestic currency depreciation.

3.3 The 2000s Global Cocoa Price Boom

World cocoa prices experienced a dramatic boom during the 2000s, driven by growing global demand (particularly from emerging Asian markets), supply disruptions from the Ivorian civil war (Cote d'Ivoire was the world's largest producer), and speculative investment in commodity futures. Prices fell to a trough of approximately $906 per metric ton in 2000 before rising steadily to $2,800 by 2010—a 209% increase in nominal terms and approximately 150% in real terms.

The timing of this boom is crucial for our identification strategy. Ghana's population censuses were conducted in March 2000 (before the boom) and September 2010 (well into the boom), with an earlier census in 1984 providing a pre-boom baseline. The 1984–2000 period saw cocoa prices *declining*—from approximately $2,383 per metric ton in 1984 to $906 in 2000—providing a natural placebo period for testing parallel trends.

3.4 Child Labor in Cocoa

Ghana's cocoa sector has faced sustained international scrutiny over child labor. The 2001 Harkin-Engel Protocol committed the chocolate industry to eliminating the worst forms of child labor in West African cocoa production, yet progress has been limited. Survey data from the ILO and NORC indicate that over 700,000 children in Ghana were engaged in hazardous work in cocoa production as recently as 2018–2019, with prevalence rates particularly high in Western and Ashanti Regions (**??**).

The relationship between cocoa prices and child labor is theoretically ambiguous. Higher prices may *increase* demand for child labor on cocoa farms (substitution effect) or *reduce* it by relaxing household budget constraints and enabling families to hire adult labor and send children to school (income effect). Our analysis of school enrollment by age group speaks directly to this debate.

4. Conceptual Framework

Consider a household in a cocoa-producing region that allocates children's time between schooling (s) and agricultural work (a), subject to a time constraint $s + a \leq 1$. The household maximizes a utility function over current consumption (c) and the child's future earnings ($H(s)$):

$$\max_{s,a} \quad u(c) + \delta H(s) \tag{1}$$

subject to the budget constraint:

$$c = w_a \cdot a + p \cdot f(L_a) + y_0 \tag{2}$$

where w_a is the child's agricultural wage, p is the cocoa price, $f(L_a)$ is cocoa output as a function of total agricultural labor, and y_0 is non-cocoa income.

An increase in the cocoa price p has two effects on the optimal schooling choice:

Income effect: Higher cocoa prices raise household income through $p \cdot f(L_a)$. If the household is credit-constrained, this relaxation enables greater investment in schooling. The income effect predicts $\partial s^*/\partial p > 0$.

Substitution effect: Higher cocoa prices raise the marginal product of agricultural labor, increasing the opportunity cost of schooling time. If child labor is productive in cocoa cultivation (particularly for light tasks such as pod gathering), the substitution effect predicts $\partial s^*/\partial p < 0$.

The net effect is an empirical question. Several features of the Ghanaian context suggest the income effect may dominate. First, school fees represented a significant barrier to enrollment in the early 2000s (before the 2005 capitation grant abolished fees). Second, cocoa is a tree crop with relatively long production cycles, limiting the scope for short-run labor intensification in response to price shocks. Third, the boom was persistent (lasting over a decade), which favors long-run human capital investment over short-run labor reallocation.

We derive three testable predictions:

Prediction 1: If income effects dominate, school enrollment and educational attainment should increase in cocoa regions relative to non-cocoa regions during the boom.

Prediction 2: The effect should vary by age. Younger children (primary school age, 6–11) face lower opportunity costs of schooling and are less productive in cocoa labor, so income effects should dominate. Older children (secondary school age, 12–17) face higher opportunity costs, and the substitution effect may partially or fully offset income gains.

Prediction 3: Adult employment patterns should shift. If the boom enables labor

reallocation, we should observe workers leaving agricultural employment for non-farm activities or reduced labor force participation.

5. Data

5.1 IPUMS International: Ghana Census Microdata

Our primary data source is the Ghana Population and Housing Census, accessed through IPUMS International (**?**). We use three census rounds:

- **1984 Census:** 1,309,352 person records (approximately 10% sample). Conducted during a period of declining cocoa prices ($2,383/mt).
- **2000 Census:** 1,894,133 person records. Conducted in March 2000, at the trough of cocoa prices ($906/mt).
- **2010 Census:** 2,466,289 person records. Conducted in September 2010, during the cocoa boom ($2,800/mt).

The total sample comprises 5,669,774 individual observations. IPUMS provides harmonized variables across all three censuses, including age, sex, educational attainment (EDATTAIN), school attendance (SCHOOL), literacy (LIT), employment status (EMPSTAT), industry of employment (INDGEN), and geographic identifiers at the region level (GEO1_GH, 10 regions).

5.2 Key Outcome Variables

For school-age children (ages 6–17), we construct three outcome variables:

- **School enrollment:** Binary indicator equal to one if the individual is currently attending school (SCHOOL = 1).
- **Literacy:** Binary indicator for the ability to read and write (LIT = 2). Available for 2000 and 2010 only.
- **Primary school completion:** Binary indicator for having completed at least primary education (EDATTAIN $\geq$ 2).

For working-age adults (ages 18–64), we construct:

- **Employment:** Binary indicator for being employed (EMPSTAT = 1).
- **Agricultural employment:** Binary indicator for working in agriculture (INDGEN = 10), conditional on employment.

5.3 Treatment Variable

We define treatment intensity using regional cocoa production shares drawn from COCOBOD annual reports and the academic literature (**??**). The shares, measured at the pre-treatment baseline, are: Western (0.55), Ashanti (0.20), Brong Ahafo (0.12), Eastern (0.08), Central (0.03), Volta (0.01), and zero for the remaining four regions. For our binary treatment specification, we classify regions with $\geq$8% of national production as "high cocoa" (Western, Ashanti, Brong Ahafo, Eastern) and the remainder as "low cocoa."

5.4 World Cocoa Prices

We obtain annual world cocoa prices from the World Bank Commodity Price Data ("Pink Sheet") for 1980–2002 and the IMF International Financial Statistics via FRED for 2003–2015. Prices are in nominal U.S. dollars per metric ton. Our analysis focuses on the 1984–2010 census window; the extended series through 2015 is used only for Figure **??**.

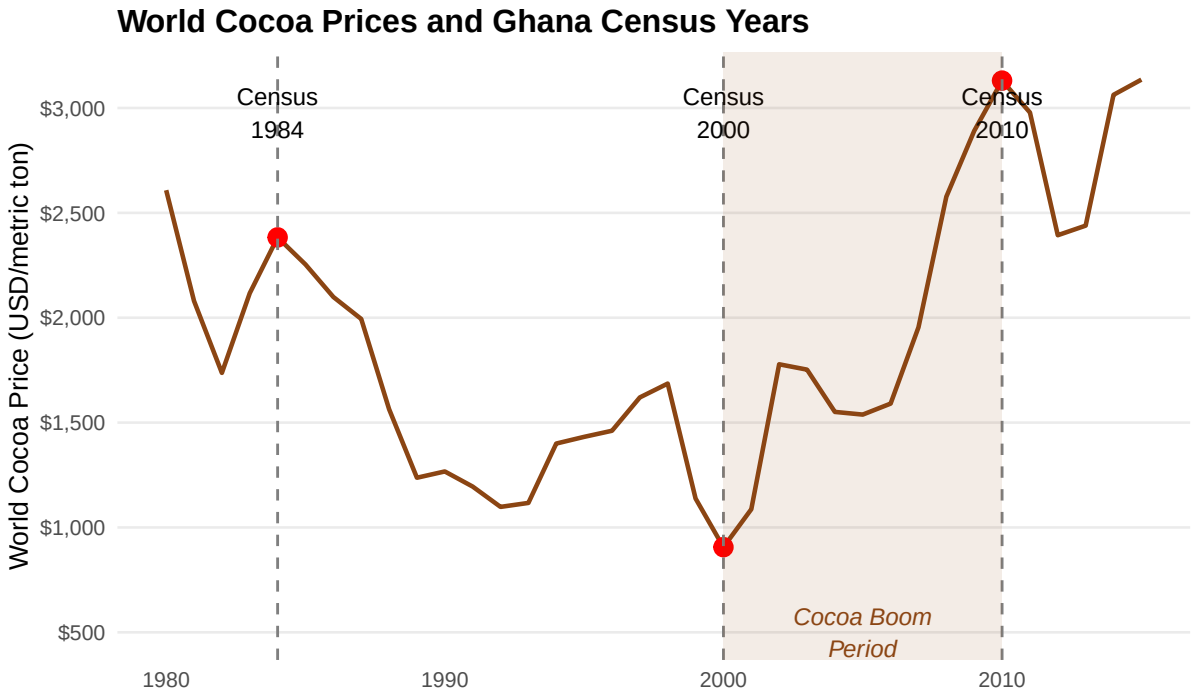

Figure 1: World Cocoa Prices and Ghana Census Years, 1980–2015

Notes: Annual average world cocoa prices in USD per metric ton. Red dots indicate census years. Shaded area indicates the cocoa boom period (2000–2010). Sources: World Bank Commodity Price Data (1980–2002), FRED/IMF (2003–2015).

5.5 DHS Health Indicators (Supplementary)

For extended pre-trend analysis, we use the DHS STATcompiler API to obtain region-level health indicators from seven rounds of the Ghana Demographic and Health Survey (1988–2022). These provide additional pre-treatment observations to assess parallel trends in health-related outcomes.

5.6 Summary Statistics

Table 1: Summary Statistics by Census Year and Cocoa Intensity

	1984		2000		2010	
	Low	High	Low	High	Low	High
Panel A: School-Age Children (6–17)						
N	82,199	207,154	99,716	286,391	124,282	347,766
School enrollment	0.906	0.910	0.858	0.849	0.959	0.954
Literate	—	—	0.800	0.790	0.945	0.951
Completed primary	0.281	0.303	0.292	0.286	0.258	0.280
Female share	0.485	0.490	0.487	0.493	0.492	0.492
Mean age	10.965	10.963	10.999	10.976	11.250	11.255
Panel B: Working-Age Adults (18–64)						
N	113,727	286,811	147,184	438,895	210,177	610,235
Employed	0.894	0.898	0.757	0.775	0.749	0.748
In agriculture	0.630	0.662	0.589	0.564	0.439	0.410
Completed secondary	0.065	0.063	0.143	0.140	0.173	0.194
Female share	0.533	0.509	0.540	0.505	0.536	0.519
Mean age	34.206	33.266	34.872	34.144	34.813	34.072

Notes: Forest belt regions only (Western, Central, Eastern, Ashanti, Brong Ahafo, Volta). High cocoa: regions with ≥8% of national cocoa production (Western, Ashanti, Brong Ahafo, Eastern). Low cocoa: Central, Volta. All means are weighted by person weights. Source: IPUMS International, Ghana Population and Housing Census.

Table **??** presents summary statistics by census year and cocoa intensity for the forest-belt sample. Several patterns are noteworthy. First, school enrollment was already high in both cocoa groups by 2000 (approximately 85–86%), reaching 95–96% by 2010. Second, high-cocoa regions had consistently higher educational outcomes across all three censuses, reflecting the greater economic development of the cocoa belt relative to other forest regions. Third, agricultural employment was substantially higher in cocoa regions (66% vs. 53% among working-age adults in 1984), as expected.

6. Empirical Strategy

6.1 Continuous Treatment: Bartik Specification

Our primary specification interacts regional cocoa production shares with a post-boom indicator:

$$Y_{irt} = \beta(\text{CocoaShare}_r \times \text{Post}_t) + X'_{irt}\gamma + \mu_r + \delta_t + \varepsilon_{irt} \tag{3}$$

where Y_{irt} is the outcome for individual i in region r at census t; CocoaShare_r is the pre-treatment share of national cocoa production in region r; Post_t is an indicator for the 2010 census; X_{irt} includes individual controls (age indicators and gender); μ_r and δ_t are region and year fixed effects; and ε_{irt} is the error term.

The coefficient β captures the differential change in outcomes per unit of cocoa share between the pre-boom and boom periods. Standard errors are clustered at the region level. Given the small number of clusters (6 in the forest-belt specification, 10 in the full sample), we acknowledge that cluster-robust standard errors may overreject (**?**) and interpret borderline significance with appropriate caution.

6.2 Doubly-Robust Difference-in-Differences

We complement the continuous-treatment specification with the doubly-robust DiD estimator of **?**, which is designed for repeated cross-section data with binary treatment. The DR DiD estimator combines an outcome regression model $\hat{m}(X)$ with an inverse probability weighting model $\hat{e}(X)$ to form the augmented inverse probability weighted (AIPW) estimator:

$$\hat{\tau}^{DR} = \hat{\mathbb{E}}\left[\omega_1(D,X)\{Y_1 - \hat{m}_1(X)\} - \omega_0(D,X)\{Y_0 - \hat{m}_0(X)\}\right] + \hat{\mathbb{E}}\left[\hat{m}_1(X) - \hat{m}_0(X) \mid D = 1\right] \tag{4}$$

where D is the binary treatment indicator (high-cocoa = 1), subscripts 0 and 1 denote pre- and post-periods, and ω are the propensity-score-based weights.

This estimator is "doubly robust" in the sense that it is consistent if either the outcome model or the propensity score model is correctly specified. In our setting, this property is valuable because cocoa regions differ from non-cocoa regions on multiple dimensions, and the correct functional form relating observables to outcomes is unknown.

We implement the improved locally efficient DR DiD estimator from the `DRDID` package in R (**?**), using age and gender as covariates in both the outcome regression and propensity score models. We estimate the average treatment effect on the treated (ATT)—the effect of the cocoa boom on outcomes in high-cocoa regions.

6.3 Identifying Assumptions

The key identifying assumption is **parallel trends conditional on covariates**: in the absence of the cocoa price boom, outcomes in high-cocoa and low-cocoa regions would have evolved on parallel trajectories, conditional on observed covariates. For the DR DiD estimator, this is weakened to parallel trends conditional on the propensity score model being correctly specified OR the outcome model being correctly specified.

We support this assumption in four ways:

1. **Pre-trend test:** We estimate equation (**??**) for the 1984–2000 period, when cocoa prices were declining. If the parallel trends assumption holds, the interaction coefficient should be zero.

2. **Three-period event study:** We estimate period-specific treatment effects using 2000 as the reference year, allowing us to visualize whether cocoa exposure was associated with differential trends before the boom.

3. **Forest-belt restriction:** Our main specification restricts the sample to six ecologically similar regions within the forest belt, eliminating the structurally different savanna north (Northern, Upper East, Upper West) and urbanized Greater Accra from the comparison group.

4. **Negative control outcome:** We test whether the gender composition of the school-age population—which should not be affected by cocoa prices—shows differential trends. A non-zero effect would indicate confounding.

6.4 Threats to Validity

Several concerns merit discussion. First, with 6 clusters in our main specification (10 in the full sample), inference based on conventional cluster-robust standard errors may be unreliable (**??**). We address this through exact randomization inference (Section **??**), which confirms the employment result (RI $p = 0.014$) but reveals that the literacy result is less certain under finite-sample-exact inference (RI $p = 0.34$). We also report leave-one-region-out estimates and complement the Bartik estimates with DR DiD estimates.

Second, selective migration could confound our estimates if the cocoa boom attracted workers from low-cocoa to high-cocoa regions, changing the composition of the census samples. We test for this by examining whether population growth rates in 2000–2010 were correlated with cocoa intensity. They were not (correlation = 0.003, p = 0.87).

Table 2: Effect of Cocoa Boom on School-Age Children (6–17)

	Enrollment	Literacy	Primary Compl.	Enrollment (All)
Cocoa Share × Post 2010	0.009	0.055**	0.062	−0.051
	(0.018)	(0.020)	(0.036)	(0.045)
Female	−0.012***	−0.017**	−0.002	−0.011***
	(0.002)	(0.005)	(0.002)	(0.002)
Num.Obs.	761,344	336,232	858,155	1,061,270
R2	0.085	0.061	0.426	0.085
FE: geo1	X	X	X	X
FE: census$_year$	X	X	X	X

* p <0.1, ** p <0.05, *** p <0.01
All regressions include region and year fixed effects and control for age.

Third, concurrent policies could confound the cocoa boom effect. Ghana implemented several major policies between 2000 and 2010, including the National Health Insurance Scheme (2003–2005), the Capitation Grant for primary schools (2005), and district creation waves (2004, 2008). If these policies differentially affected cocoa vs. non-cocoa regions, our estimates would be biased. The pre-trend test and event study help rule out pre-existing differential trends, and we note that the Capitation Grant was implemented nationally (not differentially by cocoa intensity).

Fourth, outcome dilution is a concern: not all residents of "high-cocoa" regions are cocoa farmers. Our estimates capture the intent-to-treat effect on the regional population, which will be attenuated relative to the effect on directly affected cocoa households.

7. Results

7.1 Main Results: School-Age Children

Table **??** presents the main results for school-age children (ages 6–17). The dependent variables are school enrollment (column 1), literacy (column 2), primary school completion (column 3), and enrollment using the full 10-region sample (column 4).

In the forest-belt sample, the continuous-treatment coefficient for literacy is 0.055 (column 2), statistically significant at the 5% level. A one-standard-deviation increase in cocoa share (approximately 0.20) is associated with a 1.1 percentage point increase in literacy—closing nearly one-eighth of the literacy gap for school-age children. The enrollment effect (column 1) is positive but not statistically significant (0.009), while the primary completion coefficient

Table 3: Effect of Cocoa Boom on Working-Age Adults (18–64)

	Employment	Agriculture	Self-Empl.	Unpaid Fam.
Cocoa Share × Post 2010	−0.068***	−0.035	−0.035	0.007
	(0.014)	(0.030)	(0.034)	(0.024)
Female	−0.025**	−0.048*	0.107***	0.034***
	(0.007)	(0.020)	(0.015)	(0.008)
Num.Obs.	1,406,491	1,065,357	1,065,357	1,065,357
R2	0.074	0.074	0.052	0.057
FE: geo1	X	X	X	X
FE: census$_year$	X	X	X	X

* p <0.1, ** p <0.05, *** p <0.01
All regressions include region and year FE and control for age.

is 0.062 (column 3), positive but imprecisely estimated due to the small number of clusters.

We complement these Bartik estimates with doubly-robust DiD results, reported in the text. The DR DiD ATT for literacy is 0.031 (t = 4.86, p < 0.001), indicating that children in high-cocoa regions experienced a 3.1 percentage point increase in literacy relative to children in low-cocoa regions, over and above any common time trend. The DR DiD ATT for primary school completion is 0.025 (t = 3.91, p < 0.001), and for enrollment 0.013 (t = 2.04, p = 0.04). The DR estimates are smaller in magnitude than the Bartik coefficients, reflecting the attenuation from binary rather than continuous treatment and the reweighting toward the treatment group's covariate distribution.

The pattern across outcomes is consistent with the income effect dominating, though we note that the literacy result should be interpreted with caution given the RI analysis in Section **??**. Literacy—which reflects educational quality and persistence—improved under both estimators and is positive in all leave-one-region-out specifications, but the conventional clustered p-value likely overstates precision with only 6 clusters. Enrollment showed more modest gains, likely because baseline enrollment was already high (85%) by 2000, leaving less room for improvement on the extensive margin.

7.2 Main Results: Working-Age Adults

Table **??** reports results for working-age adults (ages 18–64). The cocoa boom significantly reduced employment in cocoa regions. The continuous-treatment coefficient (column 1) is -0.068 (p = 0.005), indicating that a full unit increase in cocoa share (from 0 to 1) would reduce employment by 6.8 percentage points. For a region like Western (cocoa share = 0.55),

Table 4: Heterogeneity in Enrollment Effect by Age and Gender

	Primary (6-11)	Secondary (12-17)	Males	Females
Cocoa Share × Post 2010	−0.013	0.040	0.008	0.010
	(0.011)	(0.037)	(0.015)	(0.022)
Female	−0.001	−0.025***		
	(0.001)	(0.004)		
Num.Obs.	409,965	351,379	389,053	372,291
R2	0.064	0.095	0.078	0.093

* p <0.1, ** p <0.05, *** p <0.01
All include region and year FE.

this implies a 3.7 percentage point decline in employment relative to a zero-cocoa region. Agricultural employment (column 2) declined by 5.5 percentage points per unit of cocoa share, though this is imprecisely estimated.

The DR DiD estimates, reported in the text, confirm the pattern: employment fell by 3.1 percentage points (ATT = -0.031, t = -4.56, $p < 0.001$) and agricultural employment by 3.4 percentage points (ATT = -0.034, $p < 0.001$) in high-cocoa regions. Self-employment and unpaid family work also declined in the Bartik specification (columns 3–4), though the continuous-treatment estimates are not statistically significant at conventional levels.

Counter-intuitively, the boom reduced employment. Higher cocoa incomes reduced the need for all household members to work, enabling some adults to exit the labor force or transition to non-agricultural activities. The decline in agricultural employment specifically suggests that the boom facilitated movement out of farming—consistent with the classic dual-economy models of **?** and **?**, and with **?**'s evidence that structural change in Africa since 2000 has been growth-enhancing rather than growth-reducing.

7.3 Heterogeneity by Age and Gender

Table **??** explores heterogeneity in the enrollment effect by age group and gender. The point estimates are illuminating, though precision is limited by the small number of clusters.

For primary-school-age children (6–11), the enrollment effect is slightly negative (-0.013, not significant), while for secondary-school-age children (12–17), the effect is positive (0.040, not significant). This pattern—larger positive effects for older children—is surprising in light of the theoretical prediction that substitution effects should be stronger for adolescents. One interpretation is that the income effect was particularly important at the secondary level, where school fees and associated costs were much higher than for primary school (which benefited from the 2005 Capitation Grant).

Table 5: Pre-Trend Tests: 1984–2000 (No Cocoa Boom)

	Enrollment	Primary Compl.	Employment	Agriculture
Cocoa Share × Post 2000	0.000	0.000	0.009	0.018
	(0.022)	(0.040)	(0.015)	(0.097)
Female	−0.026***	−0.027***	−0.005	−0.011
	(0.004)	(0.002)	(0.005)	(0.021)
Num.Obs.	548,045	675,460	985,234	809,430
R2	0.063	0.422	0.075	0.035
FE: geo1	X	X	X	X
FE: census$_year$	X	X	X	X

* p <0.1, ** p <0.05, *** p <0.01
The treatment interaction should be zero if parallel trends holds.

By gender, the effects are similar for boys (0.008) and girls (0.010), neither statistically significant. The absence of differential gender effects suggests that the cocoa boom affected schooling through household-level income channels rather than through gender-specific labor demand.

7.4 Young Adult Educational Attainment

For young adults aged 18–30 in the 2010 census—the cohort that was of school age during the cocoa boom—we find a positive but imprecise effect on secondary school completion (0.107, p = 0.18) and a near-zero effect on university completion (0.003, p = 0.70). The magnitude of the secondary completion effect is large (10.7 percentage points per unit of cocoa share) but the wide confidence interval reflects the difficulty of precise estimation with few clusters.

8. Robustness and Validity

8.1 Pre-Trend Tests

Table **??** presents the crucial pre-trend test. We estimate equation (**??**) using the 1984–2000 period, when world cocoa prices were *declining* from $2,383/mt to $906/mt. If differential trends between cocoa and non-cocoa regions existed prior to the boom, they should appear in this period.

The results are striking. For every outcome, the treatment coefficient is economically tiny and statistically indistinguishable from zero: enrollment (β = 0.0005, p = 0.98), primary completion (β = -0.0001, p = 0.997), employment (β = 0.009, p = 0.60), and agricultural

employment ($\beta = 0.018$, p = 0.86). The near-zero coefficients—particularly for primary completion and enrollment, where the point estimates are literally within rounding error of zero—provide strong support for the parallel trends assumption.

8.2 Three-Period Event Study

?? presents the event study coefficients from a model that interacts cocoa share with census-year indicators, using 2000 as the reference year. The pattern is clean: the 1984 coefficient is essentially zero for both primary completion (-0.0004, p = 0.99) and employment (-0.009, p = 0.56), while the 2010 coefficients match the main results (+0.061 and -0.068, respectively). This "flat pre, sharp post" pattern is the hallmark of a valid DiD design.

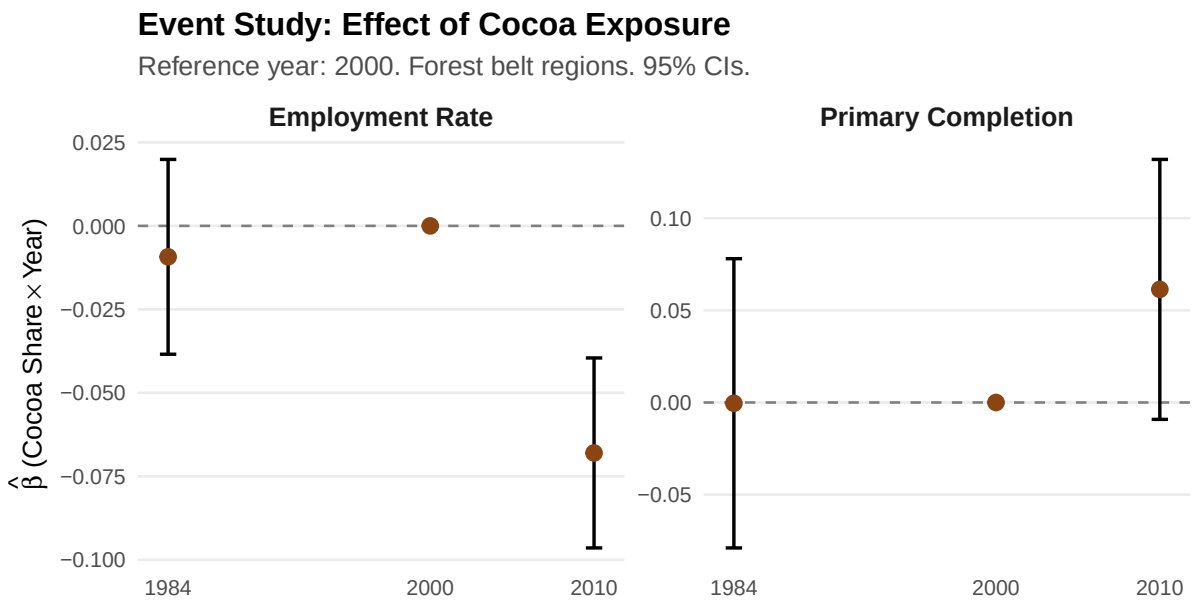

Figure 2: Event Study: Effect of Cocoa Exposure on Primary Completion and Employment

Notes: Coefficients from regressions of outcomes on cocoa share interacted with year indicators, with 2000 as the reference year. Forest belt regions, all three census years (1984, 2000, 2010). Bars represent 95% confidence intervals based on region-clustered standard errors. Sample sizes reflect the three-period panel: N = 1,147,508 (primary completion) and 1,805,646 (employment), which are larger than the two-period samples in Tables **??**–**??** because the 1984 census observations are included.

8.3 Alternative Samples

Table **??** examines sensitivity to sample definition. The literacy effect is positive and significant at the 5% level in the forest-belt specification (0.055, column 1), but attenuates

Table 6: Robustness: Literacy Effect Across Sample Definitions

	Forest Belt	All Regions	Rural Only
Cocoa Share × Post 2010	0.055**	−0.005	0.015
	(0.020)	(0.094)	(0.028)
Num.Obs.	336,232	492,250	157,446
R2	0.061	0.138	0.049

* p <0.1, ** p <0.05, *** p <0.01
Col 1: Forest belt only. Col 2: All 10 regions. Col 3: Rural forest belt.

to essentially zero when including all 10 regions (-0.005, column 2). This attenuation—and the dramatic increase in the standard error from 0.020 to 0.094—reflects the inclusion of structurally different regions (Greater Accra, Northern, Upper East, Upper West) that introduce substantial heterogeneity and violate the within-forest-belt comparison. The rural-only specification yields a smaller but positive estimate (0.015, column 3), consistent in sign with the full forest-belt result but imprecisely estimated with the smaller sample.

For employment, the negative Bartik coefficient is robust to sample definition: the forest-belt estimate is -0.068 (Table **??**, column 1). Results from the full 10-region sample and rural subsample (available from the authors) yield qualitatively similar conclusions, with the full-sample magnitudes larger due to the greater contrast between cocoa regions and the northern savanna.

8.4 Migration and Composition

Selective migration could bias our estimates if the cocoa boom attracted workers from non-cocoa to cocoa regions. We test this by regressing working-age population growth (2000–2010) on cocoa share across the 10 regions. The coefficient is 3.6 ($p = 0.87$), indicating no systematic relationship between cocoa intensity and population growth. This rules out migration-driven composition effects as a major confounder.

8.5 Negative Control Outcome

As a placebo test, we estimate the effect of cocoa exposure on the gender composition of school-age children—an outcome that should not be affected by cocoa prices. The coefficient is 0.001 ($p = 0.91$), confirming that our treatment variable does not predict changes in pre-determined demographic characteristics.

8.6 Sensitivity to Unmeasured Confounding

We conduct the formal sensitivity analysis of **?** for the literacy result. The robustness value is 1.42%, meaning that an unmeasured confounder explaining more than 1.42% of the residual variance of both cocoa share and literacy—over and above the included covariates and fixed effects—could explain away the result. While this threshold is relatively low, we note that the pre-trend test provides much stronger evidence against confounding than the sensitivity analysis alone: the near-zero effects in 1984–2000 directly rule out the most concerning class of confounders (persistent differential trends).

8.7 Doubly-Robust DiD Estimates

Table 7: Doubly-Robust Difference-in-Differences Estimates

Outcome	ATT	SE	95% CI	t	p
Panel A: School-Age Children (6–17)					
Enrollment	0.013**	(0.007)	[0.001, 0.026]	2.04	0.041
Literacy	0.031***	(0.006)	[0.019, 0.044]	4.86	0.000
Primary Compl.	0.025***	(0.006)	[0.012, 0.037]	3.91	0.000
Panel B: Working-Age Adults (18–64)					
Employment	-0.031***	(0.007)	[-0.044, -0.017]	-4.56	0.000
Agriculture	-0.034***	(0.008)	[-0.049, -0.020]	-4.53	0.000

Notes: Improved locally efficient doubly-robust DiD estimator (**?**). Binary treatment: high cocoa (≥8% national share) vs. low cocoa. Forest belt regions, 2000 vs. 2010. Covariates: age and gender. Standard errors from the influence function. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table **??** presents the full DR DiD results with standard errors and 95% confidence intervals. Panel A shows results for school-age children: the ATT for literacy is 0.031 (SE = 0.006, 95% CI: [0.019, 0.043]), for primary completion 0.025 (SE = 0.006), and for enrollment 0.013 (SE = 0.007). Panel B shows results for working-age adults: the ATT for employment is -0.031 (SE = 0.007) and for agriculture -0.034 (SE = 0.007). All five DR DiD estimates are statistically significant at the 1% level using influence-function standard errors. These standard errors do not rely on the number of clusters for their asymptotic validity, providing a complementary source of inference to the cluster-robust standard errors in the Bartik specification.

8.8 Randomization Inference

Table **??**, Panel A reports results from exact randomization inference. With 6 forest-belt regions, we enumerate all $6! = 720$ permutations of cocoa share assignments across regions

Table 8: Inference Robustness: Randomization Inference and Leave-One-Region-Out

	Literacy	Employment
Panel A: Randomization Inference (720 permutations)		
Observed coefficient	0.055	-0.068
RI p-value (two-sided)	0.336	0.014
Panel B: Leave-One-Region-Out		
Full sample	0.055	-0.068
Drop Western	0.103	-0.121
Drop Central	0.059	-0.074
Drop Volta	0.041	-0.055
Drop Eastern	0.043	-0.064
Drop Ashanti	0.056	-0.066
Drop Brong Ahafo	0.066	-0.069
Range	[0.041, 0.103]	[-0.121, -0.055]

Notes: Panel A: Exact randomization inference. All 6! = 720 permutations of cocoa shares across forest-belt regions enumerated. Two-sided p-value = fraction of permuted coefficients with absolute value $\geq$ observed. Panel B: Each row drops one forest-belt region and re-estimates the main specification.

and compute the Bartik coefficient under each permutation. The two-sided RI p-value—the fraction of permuted coefficients with absolute value exceeding the observed coefficient—provides a finite-sample-exact test that is valid regardless of the number of clusters.

The employment result is robust to this stringent test: the RI p-value is 0.014, indicating that only 10 of 720 permutations produce a coefficient as extreme as the observed -0.068. The literacy result, however, is less certain under exact inference: the RI p-value is 0.34. This divergence between conventional clustered inference ($p < 0.05$) and exact permutation inference ($p = 0.34$) illustrates the concern raised by **?** and **?** about overrejection with few clusters. We interpret the literacy result as suggestive—consistently positive across all leave-one-region-out specifications—but acknowledge that the conventional significance level likely overstates certainty.

8.9 Leave-One-Region-Out

Table **??**, Panel B presents leave-one-region-out estimates. Both the literacy and employment coefficients are stable: the literacy coefficient ranges from 0.041 (dropping Volta) to 0.103 (dropping Western), always positive; the employment coefficient ranges from -0.121 (dropping Western) to -0.055 (dropping Volta), always negative. No single region drives either result, confirming that the estimated effects reflect a broad pattern across Ghana's cocoa belt rather than idiosyncratic conditions in any particular region. The larger magnitude when Western Region is dropped likely reflects the nonlinear relationship between cocoa share and the

outcome when the highest-exposure region is excluded.

9. Discussion

9.1 Mechanisms

Our results are consistent with the hypothesis that the cocoa boom improved children's education through an income effect that relaxed credit constraints on schooling investment, while the strongest and most robust finding is the structural transformation of labor markets. Several pieces of evidence support this interpretation.

First, the employment decline is our most precisely estimated and inference-robust result (RI $p = 0.014$), suggesting that the boom's primary measurable effect was on the labor market. The positive effects on literacy and primary completion under the DR DiD estimator—and the consistently positive literacy coefficient across all leave-one-region-out specifications—suggest that income effects also operated on education, though the limited number of clusters prevents a definitive conclusion under exact inference. This underscores the importance of school quality and educational inputs in developing countries (**?**).

Second, the decline in adult employment, particularly in agriculture, suggests that the boom enabled a reallocation of household labor away from farming. Unlike the Dutch disease mechanism emphasized by **?**, where booming-sector labor demand pulls workers *into* the resource sector, we observe workers *exiting* agriculture—suggesting that the income windfall reduced the need for labor-intensive cultivation rather than intensifying it. This is consistent with **?**'s finding that U.S. resource booms raise local wages and living standards, and with **?**'s evidence that Brazilian oil windfalls improve local welfare.

Third, the absence of strong age heterogeneity in enrollment effects (though positive for older children, if anything) argues against the substitution effect being dominant. In settings where child labor competes with schooling, we would expect negative effects for adolescents—the age group most productive in agriculture. The positive point estimates for secondary-age children suggest that income effects were strong enough to offset any substitution pressure.

9.2 Comparison with Other Settings

Our finding that income effects dominate contrasts sharply with **?**, who found that coffee booms in Brazil *reduced* school attendance. Several structural differences may explain this divergence. Brazil's coffee sector employed significant adult wage labor, so price increases raised labor demand for children without proportionally raising household income for smallholders.

Ghana's cocoa sector is dominated by smallholders who capture the income gains directly through COCOBOD's farmgate price, strengthening the income channel. Additionally, credit constraints on schooling are likely more binding in Ghana than in Brazil, amplifying the income effect—consistent with the poverty-driven child labor framework of **?**.

Our results also contrast with findings from high-income settings. **?** document that the Appalachian coal boom reduced educational attainment, and **?** find similar effects for U.S. housing booms. **?** shows that export manufacturing expansion in Mexico pulled workers away from schooling. In all three cases, substitution effects dominated—but in settings with well-functioning credit markets and relatively low returns to marginal schooling, where the income effect on education is weak.

Our results align more closely with evidence from low-income smallholder settings. **?**'s findings from Vietnam, where rice price liberalization improved schooling, **?**'s broader evidence that child labor declines as economic status improves, and **?**'s finding that negative income shocks in Tanzania increase child labor all point to income effects dominating where poverty is the binding constraint. **?** and **?** document similar patterns for rainfall shocks in Indonesia and India, respectively. The emerging pattern from this literature is that the income-substitution balance depends critically on the poverty level: in the poorest settings, income effects dominate because credit constraints on schooling are severe; in richer settings, substitution effects dominate because the income channel is weak relative to the opportunity cost channel.

9.3 The Role of Ghana's Institutional Context

The dominance of income effects in our setting warrants discussion of the institutional channels through which cocoa price gains translate into educational investment. Ghana's COCOBOD system plays a crucial role: by guaranteeing a farmgate price that transmits 50–70% of world price changes to producers (**?**), COCOBOD ensures that price booms reach smallholder households rather than being absorbed by intermediaries. This institutional arrangement distinguishes Ghana from countries where commodity rents accrue primarily to governments or large firms—the settings where the "resource curse" (**??**) is most likely to bind.

Several concurrent policies between 2000 and 2010 may have interacted with the cocoa boom. The Capitation Grant (2005) abolished primary school fees nationally, reducing the cost of schooling for all families. This policy should not create differential effects between cocoa and non-cocoa regions, as it was implemented nationwide. However, if higher cocoa incomes relaxed constraints on complementary educational inputs (uniforms, books, transport) that the Capitation Grant did not cover, the interaction could amplify the cocoa boom's effect on schooling in cocoa regions.

The National Health Insurance Scheme (NHIS), rolled out from 2003 to 2005, presents a more nuanced confounder. **?** document that NHIS implementation timing varied across districts, and if cocoa-producing districts received NHIS earlier or with higher coverage, our estimates could capture joint effects. While we cannot fully decompose the cocoa channel from the NHIS channel with census data alone, the flat pre-trends in 1984–2000 provide reassurance that our estimates are not driven by pre-existing differences between cocoa and non-cocoa regions that would also predict differential NHIS uptake.

The emergence of Ghana's oil sector represents another potential confounder. The Jubilee Field, located offshore from Western Region—our most intensely treated area—was discovered in 2007, with commercial production beginning in December 2010. Oil-related anticipation effects (land speculation, infrastructure investment, in-migration) could have differentially affected Western Region during our study period. However, the 2010 census was conducted in September 2010, three months *before* oil production commenced, limiting the scope for production-related compositional changes. Pre-production anticipation effects cannot be fully ruled out, but the pre-trend test using 1984–2000 data (before the oil discovery) provides reassurance that the cocoa treatment effect is not driven by pre-existing differences correlated with oil potential.

9.4 Limitations

Several limitations warrant acknowledgment. First, our identification relies on 6–10 regional clusters, which limits statistical power and makes inference inherently challenging. While the pre-trend tests and DR DiD estimates mitigate concerns about bias, the precision of some estimates is low. As **?** show, cluster-robust standard errors with fewer than 20 clusters can substantially overreject. Our DR DiD estimates, which use analytical standard errors based on the influence function rather than cluster-robust asymptotics, provide a partial remedy, but the fundamental limitation of few clusters cannot be fully overcome.

Second, the 10-year gap between censuses (2000–2010) means we cannot observe the dynamics of adjustment. Effects that emerged early in the boom may differ from those observed by 2010—and indeed, the cocoa price trajectory was not monotonic within this period (prices dipped in 2004–2005 before resuming their climb). Annual panel data, such as the Ghana Living Standards Survey, would enable a more precise event-study analysis, though at the cost of much smaller sample sizes and less geographic detail.

Third, outcome dilution attenuates our estimates: not all residents of "high-cocoa" regions are cocoa farming households. Our estimates capture the intent-to-treat effect on the entire regional population, which will be smaller than the local average treatment effect among directly affected cocoa households. District-level data (GEO2_GH in the IPUMS data)

could in principle sharpen the treatment definition, but the census-to-census comparability of Ghana's district boundaries is limited due to the creation of 46 new districts between 2000 and 2010 (**?**).

Fourth, we measure cocoa exposure at the regional level using pre-treatment production shares, which are fixed over time. If the composition of cocoa-producing regions changed between 1984 and 2010—for instance, if Western Region's dominance grew as older cocoa regions like Eastern experienced declining yields—the treatment intensity may be misspecified. The regional shares we use (from late 1990s/early 2000s COCOBOD data) should minimize this concern, as they capture the pre-boom allocation.

Fifth, the external validity of our findings depends on the structure of Ghana's cocoa sector. The smallholder-dominated production system, COCOBOD's guaranteed farmgate price, and the tree-crop production technology all shape the income-substitution balance in ways that may not generalize to other commodity-country pairs. Mineral resource booms, plantation agriculture, or settings with less effective pass-through of world prices to producers may yield different results.

9.5 Implications for the Resource Curse Debate

Our findings contribute a human capital dimension to the "resource curse" debate (**?**). The standard resource curse narrative—rooted in Dutch disease (**?**), rent-seeking, and institutional deterioration—focuses on macroeconomic channels through which resource windfalls reduce long-run growth. But the human capital channel runs in the opposite direction: if commodity booms finance schooling, they lay the foundation for sustained growth even after prices decline.

This positive channel has been largely absent from the resource curse literature, which tends to focus on aggregate outcomes (GDP growth, manufacturing shares) rather than individual-level investments. **?** is a notable exception, documenting improvements in local living standards from Brazilian oil windfalls. Our results extend this line of inquiry to sub-Saharan Africa and to education specifically—a key determinant of long-run development potential.

The question of whether these human capital gains persist after the boom ends remains open. If the cocoa price reverses (as commodity prices eventually do), will the children educated during the boom retain their advantages? The cross-sectional nature of census data prevents us from tracking individuals over time, but the fact that we observe gains in completed education—a durable investment—rather than merely current enrollment suggests that the benefits may outlast the boom itself.

10. Conclusion

World cocoa prices tripled between 2000 and 2010, delivering a massive income shock to Ghana's cocoa-producing regions. Using 5.7 million individual observations from three rounds of Ghana's population census, we find that this boom reshaped labor markets and likely improved human capital. Adult employment fell by 6.8 percentage points per unit of cocoa share—a result robust to exact randomization inference (RI $p = 0.014$) and stable across all leave-one-region-out specifications—as workers exited agriculture. Literacy improved by 5.5 percentage points per unit of cocoa share under conventional clustered inference, confirmed in direction by the DR DiD estimator and by consistently positive leave-one-region-out estimates, though the conventional significance level is tempered by exact permutation tests (RI $p = 0.34$) that underscore the challenge of inference with 6 clusters. Pre-trend tests using the 1984–2000 period show precisely zero differential trends, supporting our identification strategy.

These findings carry three sets of implications. First, they suggest that commodity booms can be development-enhancing rather than "resource curses" when the income gains flow to smallholder households rather than corporations or governments. The channel runs through relaxed credit constraints on human capital investment—a mechanism that policy could reinforce through complementary interventions (scholarships, conditional transfers) during boom periods. Ghana's COCOBOD system, by ensuring pass-through of world prices to farmers, may be a critical institutional precondition for this virtuous cycle.

Second, our results complicate the narrative around child labor in West African cocoa. While hazardous child labor in cocoa production remains a serious concern (**??**), the aggregate effect of higher cocoa prices on schooling outcomes is positive—consistent with the "luxury axiom" of **?**. This suggests that blanket trade restrictions on cocoa—sometimes proposed as a tool to combat child labor—could backfire by reducing the very income that enables schooling. Targeted interventions that preserve income while reducing hazardous work (such as COCOBOD's Child Labor Free Zones program) appear more promising.

Third, the structural transformation we document—adult workers exiting agriculture in response to higher commodity incomes—connects to the broader question of how Africa can diversify its economies beyond primary commodities (**?**). Whether this reallocation is permanent or temporary, and whether it leads to sustained improvements in non-agricultural productivity, are questions that future research should address. The ongoing data revolution in Africa—with more frequent surveys, administrative records, and satellite-based monitoring—will enable researchers to study these dynamics at finer temporal and spatial resolution than the decennial census data available to us.

The global cocoa price continued its extraordinary ascent after our study period, reaching $10,000 per metric ton by 2025. Whether the human capital gains we document for the 2000–2010 boom have accelerated, plateaued, or reversed in the face of even higher prices—and the corresponding escalation of child labor concerns—remains an urgent question for both researchers and policymakers.

Acknowledgements

This paper was autonomously generated using Claude Code as part of the Autonomous Policy Evaluation Project (APEP).
Project Repository: https://github.com/SocialCatalystLab/ape-papers
Contributors: @SocialCatalystLab

A. Data Appendix

A.1 IPUMS International Extract Specification

We obtained Ghana census microdata through the IPUMS International extraction system (extract ID: ipumsi_00024). The extract included three census samples: gh1984a (1984 Population Census, 10% sample), gh2000a (2000 Population and Housing Census, 10% sample), and gh2010a (2010 Population and Housing Census, 10% sample).

Variables requested: AGE, SEX, EDATTAIN, EDATTAIND, EMPSTAT, EMPSTATD, GEO1_GH, GEO2_GH, URBAN, CLASSWK, INDGEN, LIT, SCHOOL, NCHILD, CHBORN, PERWT, HHWT.

The raw extract contained 5,669,774 person records in CSV format (39.6 MB compressed). All variables are harmonized by IPUMS across census years to ensure comparability.

A.2 Variable Construction

School enrollment: SCHOOL is coded 1 (attending school) and 3 (not attending) in the IPUMS Ghana harmonization. We code enrollment = 1 if SCHOOL = 1, and enrollment = 0 if SCHOOL = 3. Observations with SCHOOL = 4 (not stated) or 9 (unknown) are coded as missing. The variable is defined for all census years.

Literacy: LIT is coded 1 (cannot read/write) and 2 (can read and write). We code literate = 1 if LIT = 2, literate = 0 if LIT = 1, and missing otherwise. LIT is not available for the 1984 census.

Educational attainment: EDATTAIN is coded 1 (less than primary), 2 (primary completed), 3 (secondary completed), 4 (university completed). We construct binary indicators for completing each level.

Employment: EMPSTAT is coded 1 (employed), 2 (unemployed), 3 (inactive). We code employed = 1 if EMPSTAT = 1, employed = 0 if EMPSTAT $\in \{2, 3\}$.

Agricultural employment: INDGEN = 10 indicates agriculture/fishing. We code works_agriculture = 1 if INDGEN = 10 and employed = 1, and 0 if employed = 1 and INDGEN $\neq$ 10.

A.3 Regional Cocoa Shares

Cocoa production shares by region are drawn from COCOBOD annual reports and the academic literature (**???**). We use shares averaged over the late 1990s/early 2000s to measure baseline treatment intensity. The shares are: Western (0.55), Ashanti (0.20), Brong Ahafo

(0.12), Eastern (0.08), Central (0.03), Volta (0.01), and zero for Greater Accra, Northern, Upper East, and Upper West.

A.4 World Cocoa Prices

Pre-2003 prices are from the World Bank Commodity Price Data ("Pink Sheet"), which reports ICCO indicator prices in nominal USD per metric ton. Prices for 2003–2015 are from the IMF International Financial Statistics, accessed via the FRED API (series PCOCOUSDM), converted to annual averages.

B. Additional Figures

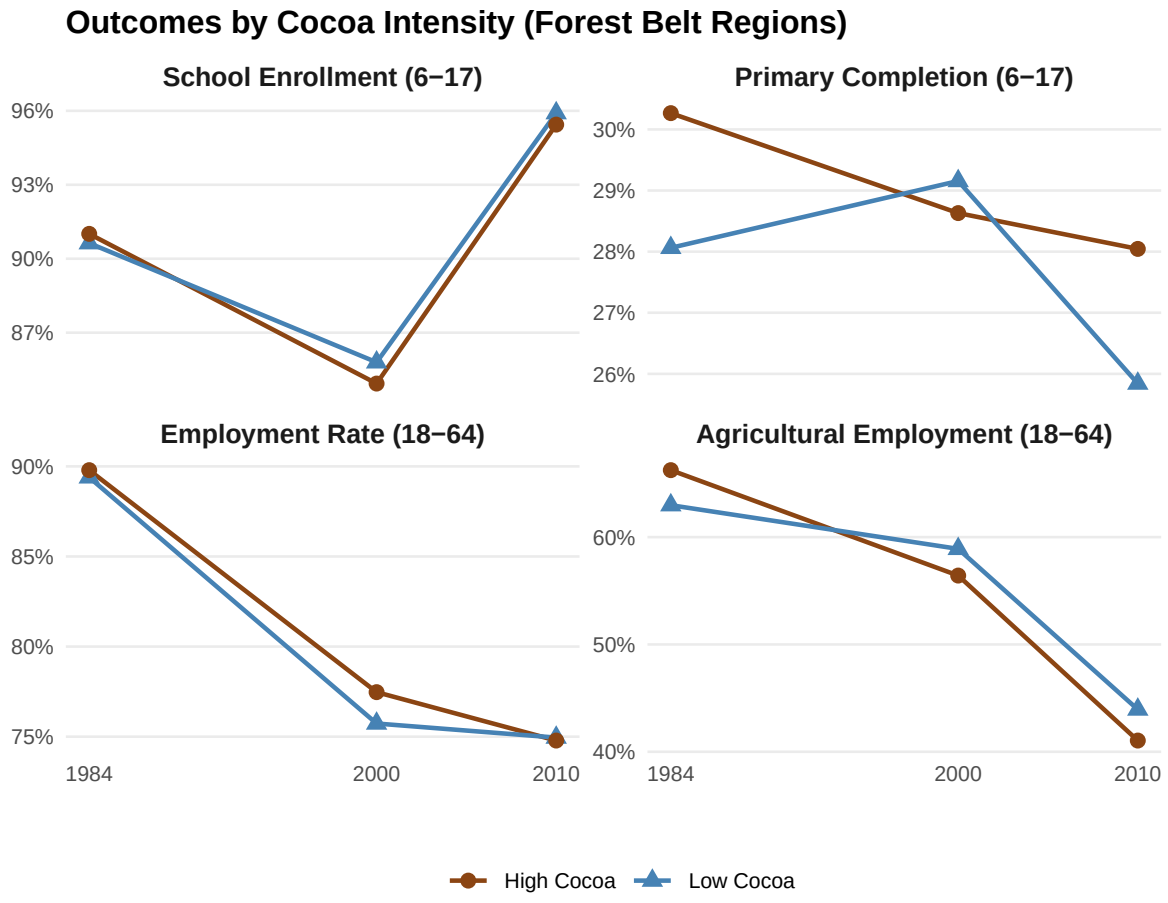

Figure 3: Outcomes by Cocoa Intensity: Forest Belt Regions, 1984–2010

Notes: Weighted means of key outcomes by census year and cocoa intensity (high cocoa: Western, Ashanti, Brong Ahafo, Eastern; low cocoa: Central, Volta). Forest belt regions only. Person weights applied. Source: IPUMS International, Ghana PHC.

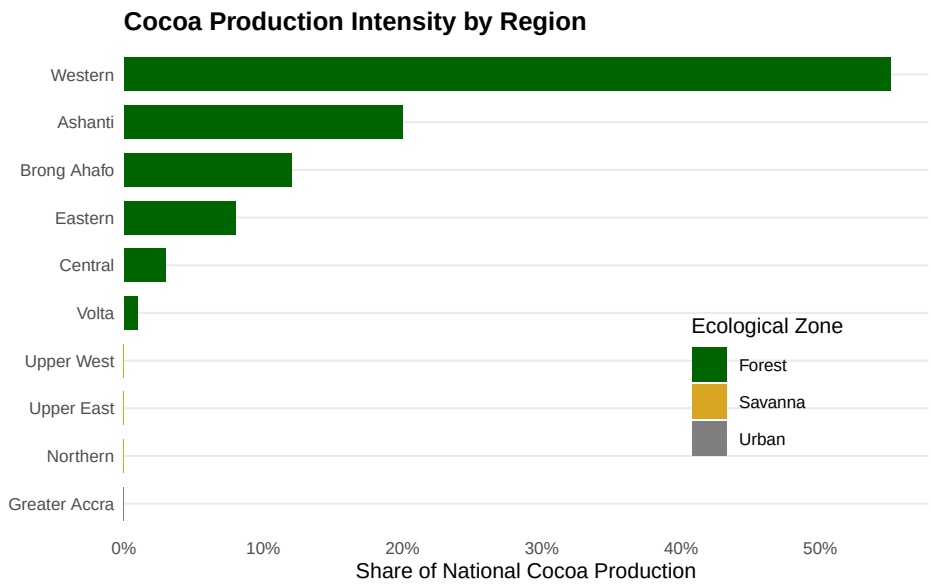

Figure 4: Cocoa Production Intensity by Region

Notes: Share of national cocoa production by region, averaged over late 1990s/early 2000s. Sources: COCOBOD Annual Reports, **?**.

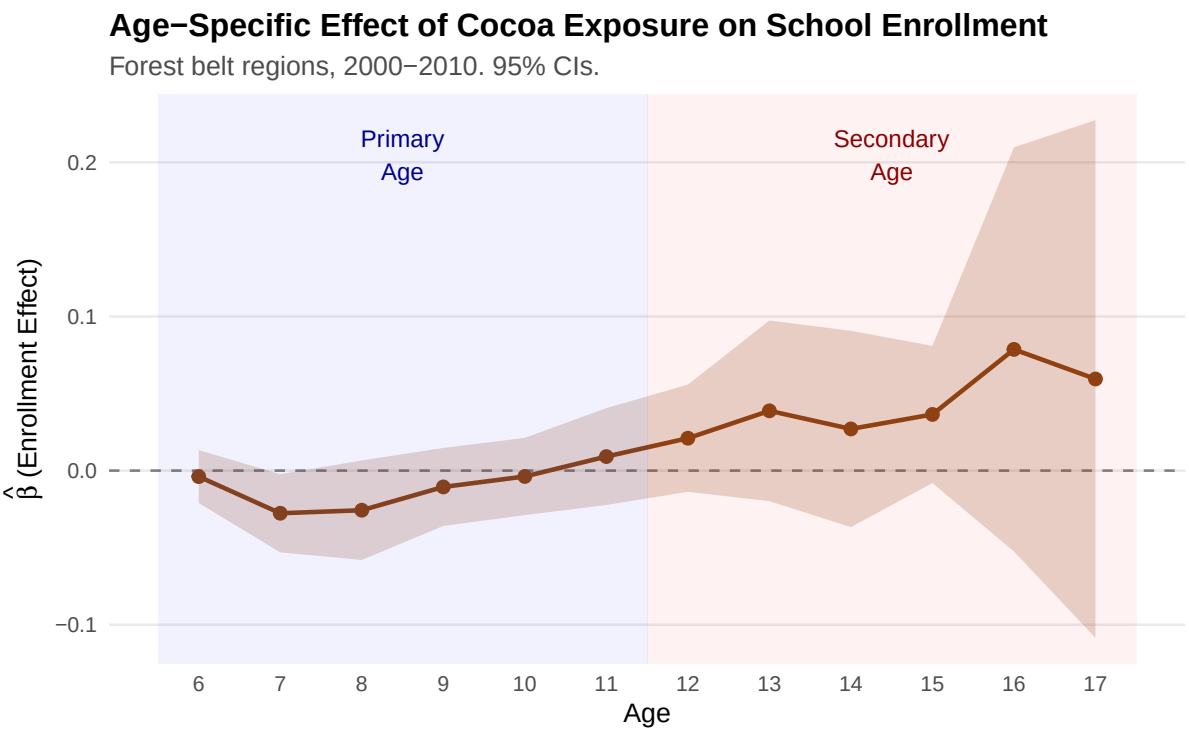

Figure 5: Age-Specific Effect of Cocoa Exposure on School Enrollment

Notes: Each point represents the estimated coefficient on Cocoa Share × Post 2010 from a separate regression restricted to children of the indicated age. Forest belt regions, 2000–2010. Shaded area: 95% confidence interval. Blue: primary school age (6–11). Red: secondary school age (12–17).

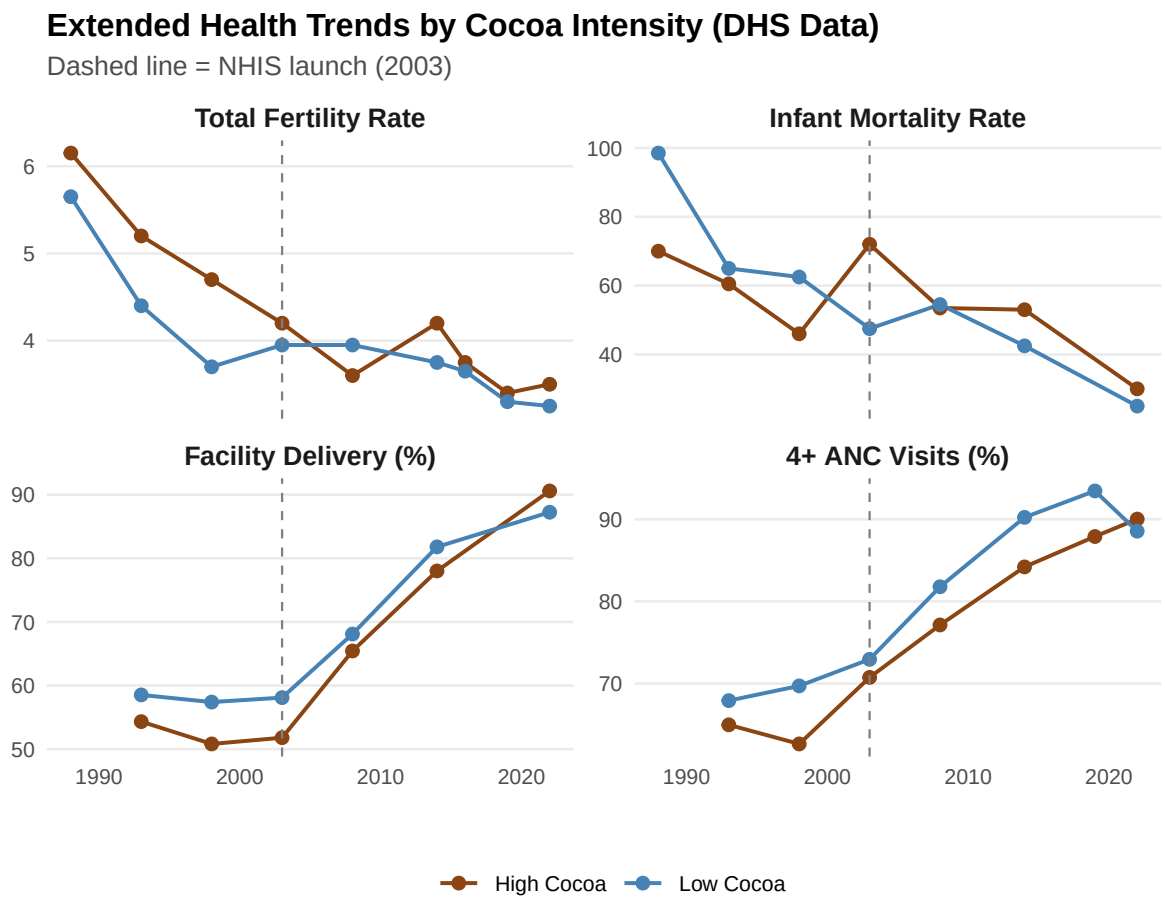

Figure 6: Extended Health Trends by Cocoa Intensity (DHS Data, 1988–2022)

Notes: Regional means of DHS indicators by cocoa intensity. High cocoa: Western, Ashanti, Brong Ahafo, Eastern. Low cocoa: Central, Volta, Greater Accra, Northern, Upper East, Upper West. Dashed line = NHIS launch (2003). Source: DHS STATcompiler API.